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Linear Programming Assignment: Optimal Number of Flights for Two Aircraft

MATS JORIS (NIU: 1826859), TODO: Author 2 (add name and NIU), and TODO: Author 3 (add name and NIU)
Master on Research and Innovation in Computer based Science and Engineering, Universitat Autònoma de Barcelona, Spain

ABSTRACT An airline has two aircraft and wants to know how many flights each one should make. We model this as a linear programming problem and solve it with the graphical method and with the simplex method in MATLAB. The maximum benefit is 190 000 euro with 160 flights of A and 40 flights of B. The minimum fuel use is reached with 30 flights of each aircraft.

INDEX TERMS Graphical method, linear programming, MATLAB, simplex method.

I. INTRODUCTION

An airline has two aircraft, A and B, for one route. We want to know how many flights each aircraft should make to earn the most money. First, we write the problem as a linear programming (LP) problem. Then we solve it with the graphical method (Part 1a), check if a constraint can be removed (Part 1b) and solve it again with the simplex method in MATLAB (Part 1c). Finally, in Part 2 we look for the number of flights that uses the least fuel.

II. PROBLEM FORMULATION

The flights must follow these rules: aircraft A must fly more times than aircraft B (C1), but at most 120 flights more (C2); aircraft B can make at most 180 flights (C3); and together the two aircraft must make at least 60 flights (C4) and at most 200 flights (C5). The company earns 1000 € on each flight of A and 750 € on each flight of B.

To write this as an LP problem, let x_1 be the number of flights of aircraft A and x_2 the number of flights of aircraft B. We want the highest benefit, so the objective function is

$$\max Z = 1000\,x_1 + 750\,x_2. \tag{1}$$

The constraints are:

$$\begin{aligned} x_1 &\geq x_2 && \text{(C1)}\\ x_1 - x_2 &\leq 120 && \text{(C2)}\\ x_2 &\leq 180 && \text{(C3)}\\ x_1 + x_2 &\geq 60 && \text{(C4)}\\ x_1 + x_2 &\leq 200 && \text{(C5)}\\ x_1,\ x_2 &\geq 0 && \text{(nonnegativity)} \end{aligned}$$

For C1 we use $x_1 \geq x_2$ instead of $x_1 > x_2$, because LP does not work with strict inequalities.

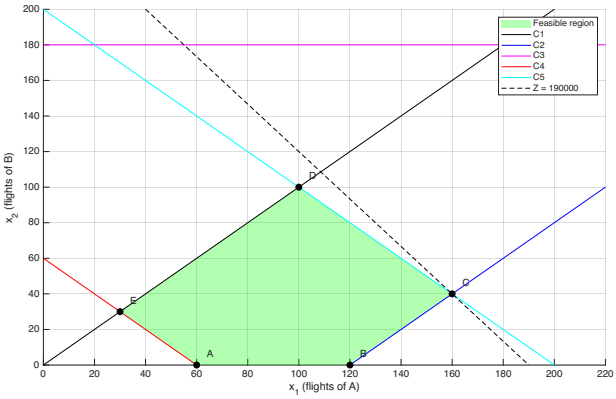


FIGURE 1. Graphical solution. The dashed line is the isoprofit line through the optimum.

III. PART 1A: GRAPHICAL SOLUTION

First, we solve the problem with the graphical method. We draw the line of each constraint in MATLAB (Fig. 1). The green area is the feasible region: all points that satisfy every constraint. The optimal solution is always in an extreme point (corner) of this region, so we compute Z in each extreme point (Table 1).

TABLE 1. Extreme points and value of Z.

Point	Intersection	(x_1, x_2)	Z (€)
A	C4 and $x_2 = 0$	$(60, 0)$	60 000
B	C2 and $x_2 = 0$	$(120, 0)$	120 000
C	C2 and C5	$(160, 40)$	**190 000**
D	C1 and C5	$(100, 100)$	175 000
E	C1 and C4	$(30, 30)$	52 500

Z is highest in point C. So the best solution is $x_1 = 160$

flights for A and $x_2 = 40$ flights for B, which gives $Z = 190\,000$ €.

IV. PART 1B: REDUNDANT CONSTRAINTS

Next, we check if a constraint can be removed without changing the solution. The answer is yes: constraint C3 ($x_2 \leq 180$) is a redundant constraint, so it can be removed. From C1 and C5 we get $2x_2 \leq x_1 + x_2 \leq 200$, so $x_2 \leq 100$ in the whole feasible region. In Fig. 1 the line of C3 lies above the green area. Without C3, the feasible region and the solution stay the same.

C1 and C4 can also be removed. Point C lies only on the lines of C2 and C5, so without C1 or C4 the feasible region gets bigger, but point C is still the best point. Only C2 and C5 cannot be removed, because they define point C.

V. PART 1C: SIMPLEX METHOD WITH MATLAB

Now we solve the same problem with the simplex method, using the MATLAB solver `linprog`. This solver finds the minimum of $f^T x$ subject to $Ax \leq b$ and $x \geq lb$. Our problem is a maximization, so we minimize $-Z = -1000\,x_1 - 750\,x_2$ instead, which gives $f = [-1000 \;\; -750]^T$. The nonnegativity constraints give $lb = [0 \;\; 0]^T$. The other constraints must be written as $\leq$, so we multiply C1 and C4 by -1. This gives:

$$A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \\ 0 & 1 \\ -1 & -1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 120 \\ 180 \\ -60 \\ 200 \end{bmatrix}.$$

We choose the simplex algorithm of `linprog`:

```
options = optimoptions('linprog', ...
          'Algorithm','dual-simplex');
[x,fval] = linprog(f,A,b,[],[],lb,[],options)
Z = -fval
```

MATLAB gives:

```
Optimal solution found.
x =
   160
    40
fval =
     -190000
Z =
      190000
```

So $x_1 = 160$, $x_2 = 40$ and $Z = -\text{fval} = 190\,000$ €. This is the same result as with the graphical method (point C).

VI. PART 2: MINIMUM FUEL CONSUMPTION

In Part 2 the goal changes: we want the lowest total fuel consumption instead of the highest benefit. Aircraft A uses twice as much fuel per flight as aircraft B, so if one flight of B uses 1 unit of fuel, one flight of A uses 2 units. The constraints stay the same, and we minimize the total fuel:

$$\min F = 2\,x_1 + x_2.$$

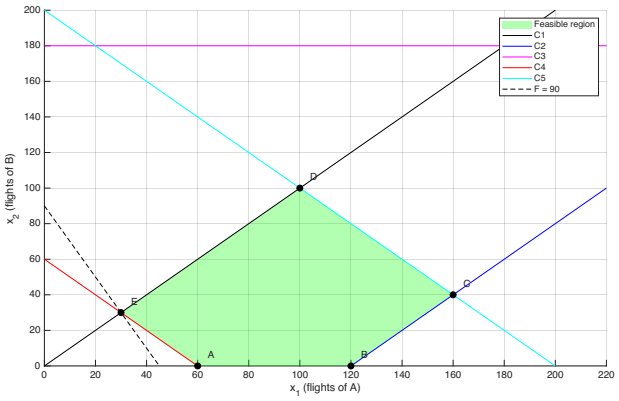


FIGURE 2. Graphical solution of Part 2. The dashed line is the isocost line through the optimum.

Graphical method. The feasible region is the same as in Part 1 (Fig. 2), so we compute F in the same extreme points (Table 2).

TABLE 2. Extreme points and value of F.

Point	(x_1, x_2)	F (units)
A	$(60, 0)$	120
B	$(120, 0)$	240
C	$(160, 40)$	360
D	$(100, 100)$	300
E	$(30, 30)$	**90**

F is lowest in point E, so $x_1 = x_2 = 30$ and $F = 90$ units.

Simplex method with MATLAB. We use the same A, b and lb as in Part 1c, but now with $f = [2 \;\; 1]^T$. This problem is already a minimization, so we do not change the sign. MATLAB gives:

```
Optimal solution found.
x =
    30
    30
fval =
    90
```

This is the same result as with the graphical method.

Note. In point E both aircraft make 30 flights. C1 allows this, but in the problem A must fly more times than B. If A must make at least one flight more ($x_1 \geq x_2 + 1$), the best whole numbers are $x_1 = 31$ and $x_2 = 29$, with $F = 91$ units.

VII. CONCLUSION

In Part 1, the maximum benefit is $Z = 190\,000$ € with 160 flights of A and 40 flights of B. Both methods give the same result, and constraint C3 is redundant. In Part 2, the minimum fuel consumption is $F = 90$ units with 30 flights of each aircraft (or $F = 91$ with 31 and 29 flights if A must fly strictly more than B).